Learning Complex Physical Systems

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AI in Industry
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Overview

1. Background and Motivation
2. Data
   - Generation
   - Inspection
   - Preprocessing

Double Pendulum

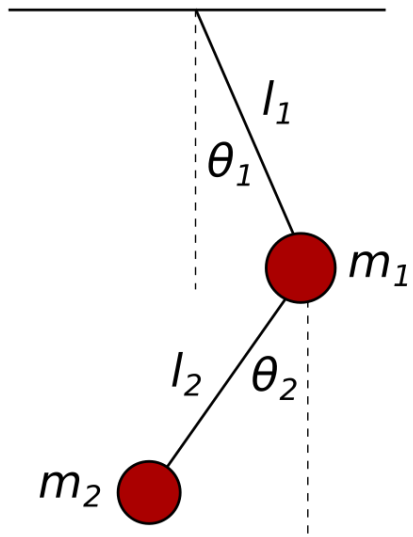


Figure: Double Pendulum.

For the sake of simplicity, we refer to the state as $q = [\theta_1, \theta_2]^T$.

Double Pendulum Equations

Inertia Matrix $M(q)$

$$\begin{bmatrix} (m_1+m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 \cos(q_2) & m_2 l_2^2 + m_2 l_1 l_2 \cos(q_2) \\ m_2 l_2^2 + m_2 l_1 l_2 \cos(q_2) & m_2 l_2^2 \end{bmatrix} \tag{1}$$

Gravity Vector $G(q)$

$$G(q) = \begin{bmatrix} (m_1+m_2) g l_1 \sin(q_1) + m_2 g l_2 \sin(q_1+q_2) \\ m_2 g l_2 \sin(q_1+q_2) \end{bmatrix} \tag{2}$$

Damping D

$$D = \begin{bmatrix} b_1 & 0 \\ 0 & b_2 \end{bmatrix} \tag{3}$$

Coriolis Vector

$$\mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -m_2 l_1 l_2 s_2 (2\dot{\theta}_1 + \dot{\theta}_2)\dot{\theta}_2 \\ m_2 l_1 l_2 s_2 \dot{\theta}_1^2 \end{bmatrix} \tag{4}$$

Dynamics Equation

$$M(q)\ddot{q} + \mathbf{c}(q, \dot{q}) + D\dot{q} + G(q) = u \tag{5}$$

Motivation

- Real physical systems are chaotic and non-linear
- Parameter estimation with sensor is expensive and noisy
- ...
- ...

Data Generation

Simulation Parameters

- **Time:** 10 s duration with $\Delta t = 0.01$ s step.
- **Initialization:** $q_0 \sim \mathcal{U}[-\pi, \pi]$ and $\dot{q}_0 = 0$.
- **System:** Masses (m_i) and lengths (l_i) are fixed arbitrarily.

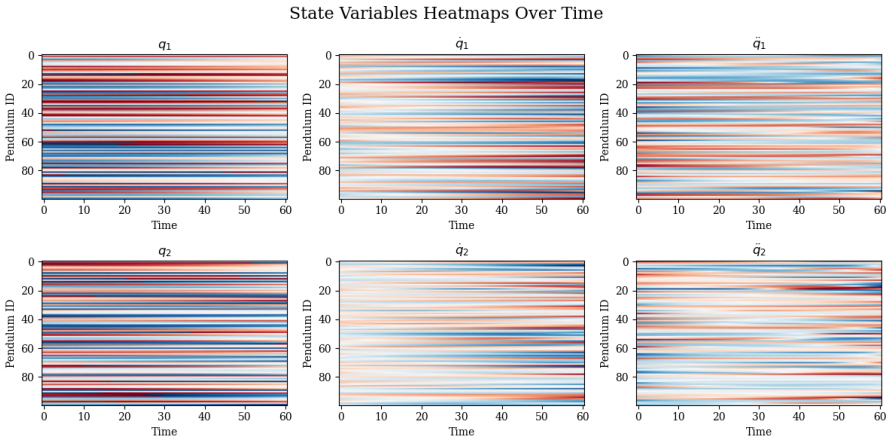
Generated Datasets

High Friction
$b = 1.0$

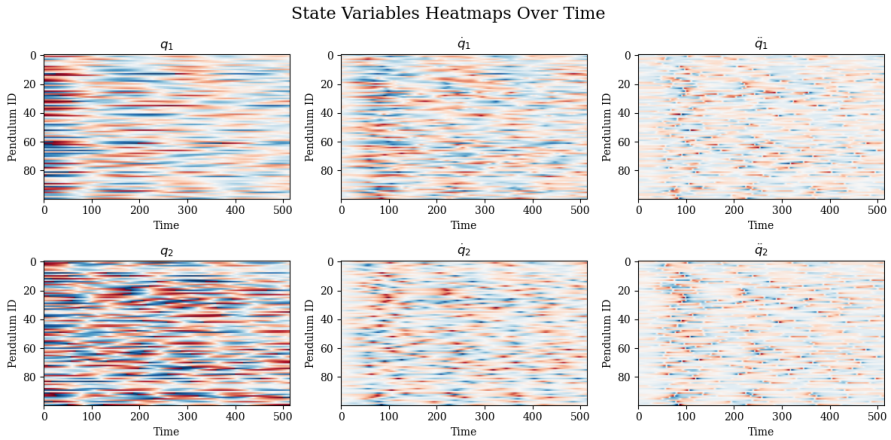
Low Friction
$b = 0.3$

No Friction
$b = 0.0$

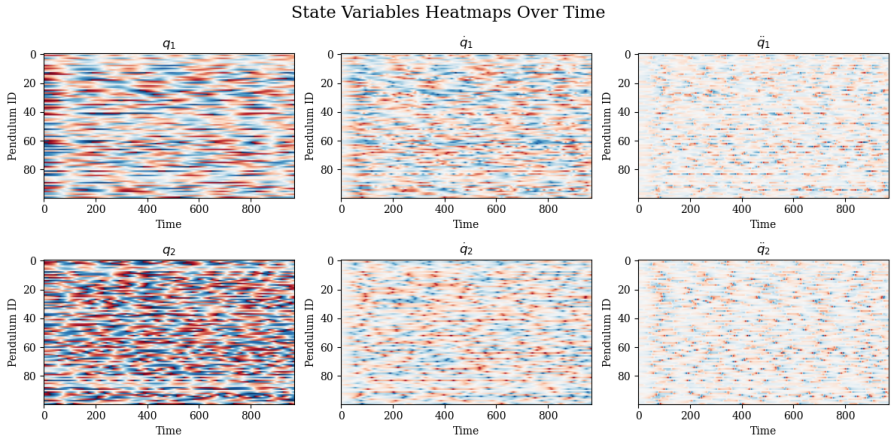
Data Inspection: Heatmap



High Friction

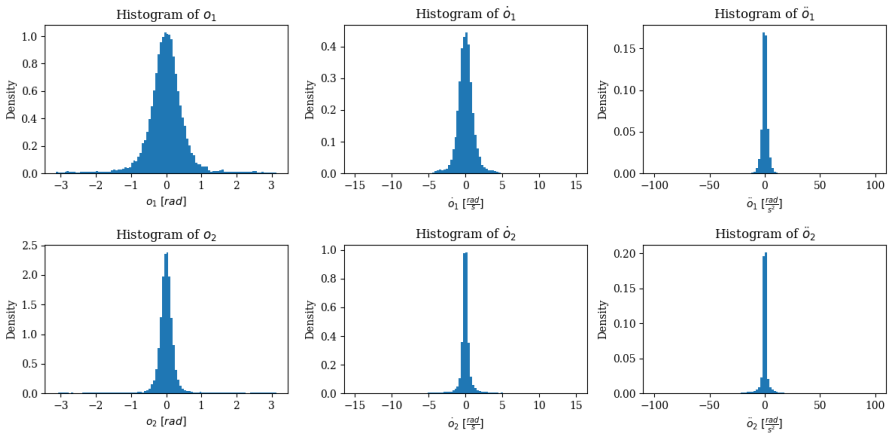


Low Friction

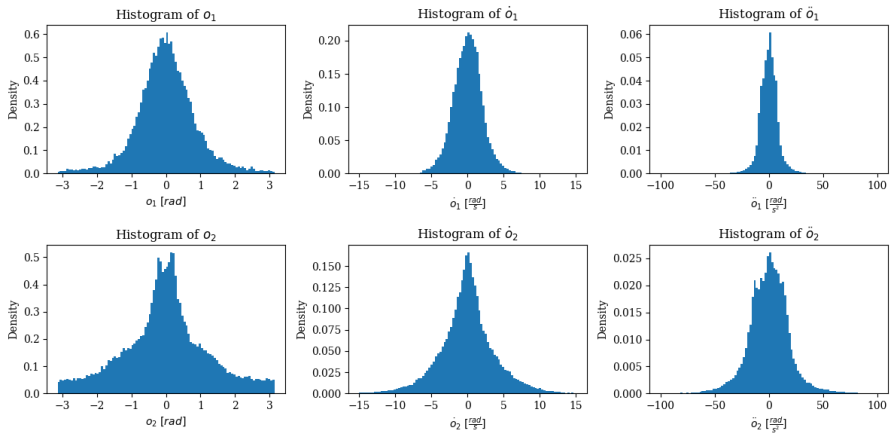


No Friction

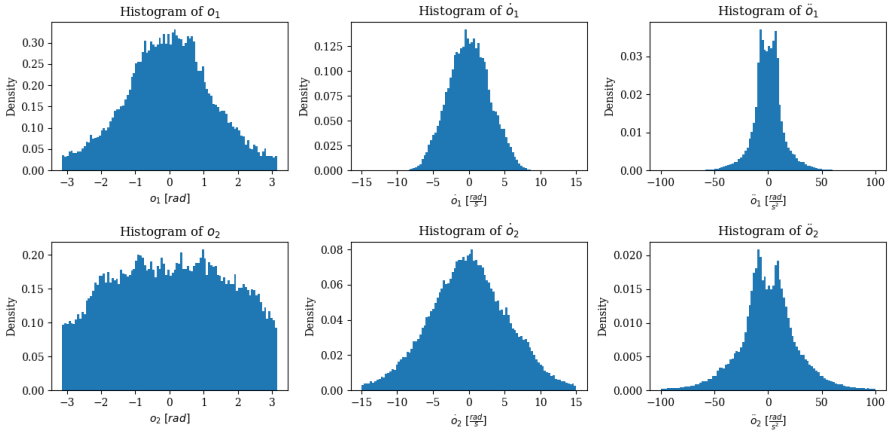
Data Inspection: State Distribution



High Friction

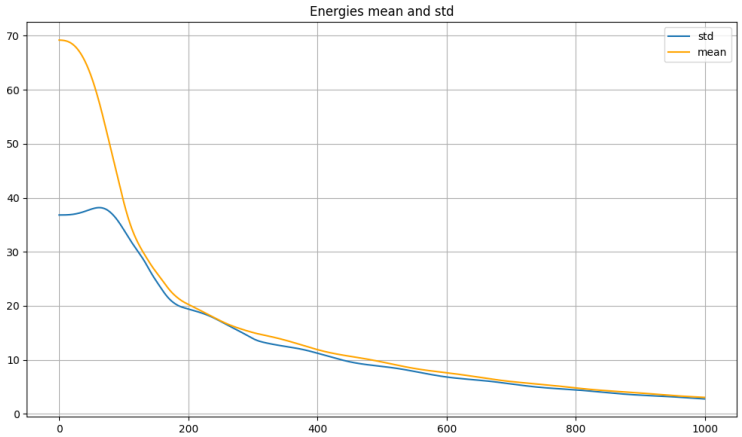


Low Friction

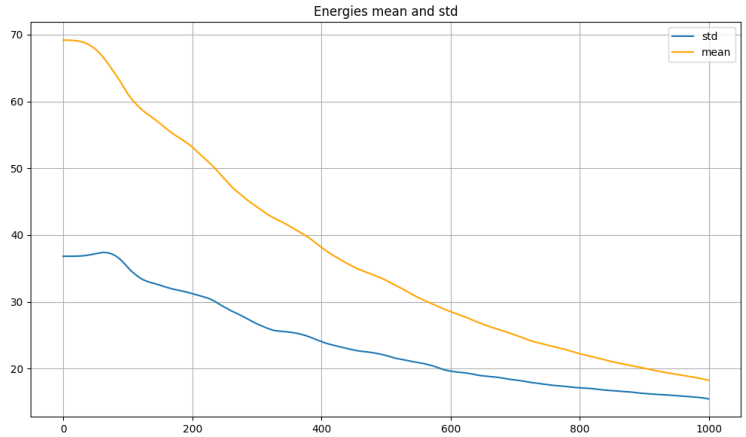


No Friction

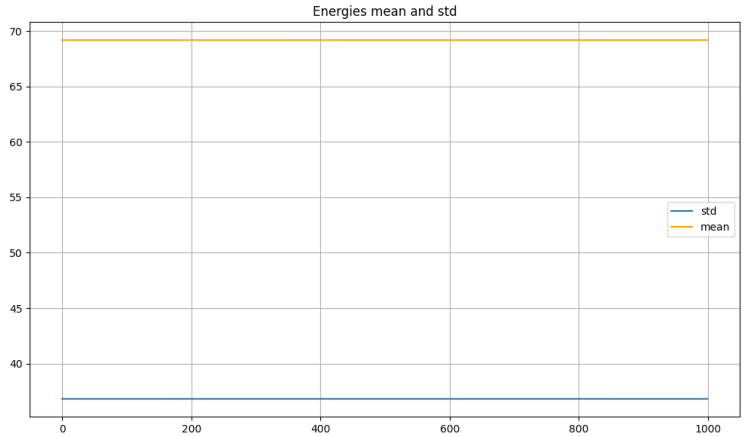
Data Inspection: Energy Statistics



High Friction



Low Friction



No Friction

Data Preprocessing Pipeline

- **Energy Conservation Check:** discard entire simulations where energy deviation exceeds tolerance ($\Delta E > 10^{-2}$).
- **Dynamic Time Cut:** truncate simulation duration T to maximize uniformity of state distribution (via KL Divergence).

$$T^* = \arg\min_T D_{KL}(P_{\text{data}} || U_{\text{uniform}})$$

- **Noise Injection:** add Gaussian noise $\mathcal{N}(0, 0.01)$ to states.
- **Angle Wrapping:** map angles to $[-\pi, \pi]$ via modulo operation.